

Sarang Vehale

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Professional Summary

Cybersecurity and quantum computing researcher with experience in OSINT systems, law enforcement-focused AI pipelines, and quantum algorithm development. Work spans cybercrime research initiatives, quantum cryptography, hybrid algorithms, and many-body systems, along with open-source and systems-level implementations.

Education

Dual degree pursuit in Cyber Security and Data Science from NFSU and IITM.

National Forensic Sciences University

Delhi, India

B.Tech - M.Tech in Cyber Security

GPA: 9.20/10.0

May 2026

Relevant Coursework: Assembly, C/C++, Digital Forensics, Cyber Security, AI & ML, Network and Mobile Security

Indian Institute of Technology, Madras

Chennai, India

B.S in Data Science

GPA: 6.95/10.0

May 2027

Relevant Coursework: Artificial Intelligence, Calculus, Data Analysis, Machine Learning

Gate Qualified

Indian Institute of Technology, Guwahati

Feb 2026

Stream: Computer Science

Experience

Center for Development of Advanced Computing (CDAC)

Pune, India

Research Intern

August 2025 - March 2026

- Developed a benchmarking framework for quantum computers, utilizing exact diagonalization algorithms (CERTIFY-ED) for system validation.
- Delivered exact results for benchmarking Noisy Low-Scale Quantum Systems (NLSQ), contributing to the evaluation of quantum computational capabilities.

National Cyber-Crime Research & Innovation Centre (NCR&IC), I4C, MHA

Delhi, India

Security Researcher and Mentor

December 2024 – March 2025

- Developed a flagship OSINT platform, integrated into the Swayam platform for Law Enforcement Agencies (LEAs).
- Conducted specialized training sessions for judicial officers and LEAs on cybersecurity best practices and investigation techniques.
- Coordinated the Cyber Commandos Program, enhancing response strategies for cybersecurity threats.
- Trained large language models (LLMs) for NLP applications, tailored to LEA-specific requirements and challenges.

Centre for Development of Advanced Computing (CDAC)

Pune, India

Research Intern

May 2024 – July 2024

- Designed hybrid quantum algorithms combining Babai's and Schnorr's algorithms with Quantum Approximate Optimization Algorithm (QAOA) for enhanced computational efficiency.
- Authored a research paper on integer factorization using lattice-based quantum algorithms, contributing to quantum cryptography advancements.

Quantum Open Source Foundation (QOSF)

Canberra, Australia

Research Intern

April 2024 – June 2024

- Investigated advanced key distribution protocols employing quantum cryptographic techniques to improve secure communications.
- Proposed a novel framework for utilizing quantum teleportation in interstellar communication, advancing quantum networking protocols.

Stanford Code in Place

Remote

Section Leader

April 2025 – May 2025

- Led weekly sections for CS106A: Introduction to Python Programming, supporting students in mastering Python using the Karel pedagogy tool.
- Facilitated integration of large language models (LLMs) into course materials, helping students explore AI applications in Python programming.
- Collaborated with Stanford instructors to design interactive activities and assignments, enhancing the learning experience for a global audience.

Bhoomi-Itech

Delhi, India

Research Intern

May 2023 – July 2023

- Established CALDERA and MITRE ATT&CK frameworks for threat simulations, improving the lab's cybersecurity capabilities.
- Contributed to triaging and analyzing simulated cybersecurity incidents, identifying vulnerabilities, and strengthening response strategies.

Research & Publications

- Some Results on Cumulative Residual Inaccuracy Measure of k-Record Values (MDPI Entropy)
- Hierarchical quantum accelerated federated learning for scalable auditable cross enterprise AI governance (IJSET)
- Survey of Integer Factorization using lattice based algorithms (Springer Nature, SN Computer Science)
- Book Chapter - From Qubits to Quantum Radar: A Comprehensive Exploration of Quantum Machine Learning (CRC press)

Research Projects

- **CERTIFY-ED** – *Formal Validation and Benchmarking Framework for Exact Diagonalization of Quantum Many-Body Systems* (under publication, MDPI Axioms)
 - Designed a multi-oracle validation framework integrating SageMath, NumPy, and Lanczos methods for cross-verification of quantum simulations
 - Implemented symbolic Hermiticity certification and deterministic eigenvalue resolution to ensure reproducibility
 - Achieved $< 10^{-11}$ relative error and 12-digit agreement with analytical solutions across benchmark systems
- **EDforManybody** – *Exact Diagonalization Engine for Quantum Many-Body Hamiltonians*
 - Developed a scalable exact diagonalization framework for analyzing quantum many-body systems
 - Optimized matrix diagonalization and memory handling to extend simulation capacity for larger Hilbert spaces
 - Used as a benchmarking backbone for validating quantum algorithms and numerical methods
- **Integer Factorization using Lattice-Based Methods (CDAC Internship Work)** – *Hybrid Quantum-Classical Approaches to Cryptographic Problems*
 - Investigated lattice-based integer factorization techniques combining Babai's and Schnorr's algorithms with QAOA

- Evaluated feasibility of hybrid quantum approaches for cryptographic applications and post-quantum security
- Contributed to research on improving efficiency of factorization under constrained quantum resources
- **Quantum Cryptographic Protocols (QOSF Internship Work)** – *Teleportation-Based Secure Communication Systems*
 - Developed quantum communication protocols leveraging teleportation for secure information transfer
 - Modeled effects of noise and decoherence on protocol reliability in long-distance communication
 - Explored applications in quantum networking and deep-space communication scenarios

Dissertation - NFSU

- **Exact Diagonalization of Quantum Many-Body Hamiltonians**
 - Developed exact solutions for benchmark quantum many-body systems to validate approximate numerical methods
 - Analyzed spectral properties and system behavior under varying interaction models
- **Optimizing Quantum Teleportation via Convex Programming**
 - Formulated quantum teleportation as a semidefinite optimization problem using information-theoretic constraints
 - Achieved up to 15
 - Introduced entropy-based regularization to enforce physically valid quantum states
- **CERTIFY-ED: Formal Verification for Exact Diagonalization**
 - Developed a symbolic and numerical validation pipeline for quantum simulations using multi-implementation consensus
 - Established reproducibility guarantees across heterogeneous computational environments

Projects

Quantum & Systems

- **CERTIFY-ED** (Python) - Proposed framework for validation framework
- **EDforManybody** (Python) - Benchmark ED scaling for different models using exact diagonalization as the ED engine.

Developer Tools

- **dotman** (Bash, Python) - A modular, TUI-based, Git-powered dotfiles manager
- **rdd** – Rust reimplement of GNU dd featuring memory-safe design, extensible I/O architecture, and integrated cryptographic verification (BLAKE3/SHA-256)
- **dotatui** (Rust) - A terminal-based Git client written in Rust, inspired by ‘lazygit’ and the UI philosophy of ‘lazyvim’.
- **dance-linux** (Python) - An interactive linux learning, CLI-based tool across multiple levels.

Security & Data

- **reconGram** (Python) - Instagram OSINT Tool
- **map-this** (Python) - Dynamic, interactive, localized crime map based on data from cybercrime.gov.in
- **personal-document-chatbot** (Python) - Simple implementation of Ollama with Langchain to develop a personalized chatbot with Q&A and chat history.

Applications

- **SimpleThoughts** (Typescript) - Minimal journal app
- **finance-footprint** (Typescript) - Finance tracking application with insights and spending analysis.
- **Yanktube** (React/Python) - Youtube video and mp3 downloader
- **chatty-docs-helper** (Typescript/Python) - Forensic document analysis using LLMs

Web & Portfolio

- **Portfolio Website (Client Project)** – Built using HTML, CSS, Bootstrap, and JavaScript
- **Legacy Portfolio Website** – Dynamic portfolio, later replaced for maintainability
- **Current Portfolio (Hugo)** – Static site using Hugo for performance and maintainability

Flutter - experimental/learning

Weather App . **Assent Application** . **Calculator** . **Cryptocurrency trading App**

Infrastructure & Homelab

- Built and maintain a Proxmox-based homelab environment
- Deployed Plex, Nextcloud, and Pi-hole across virtual machines
- Experience configuring Linux systems, storage, and service orchestration
- Worked with firewall rules, system hardening, and access control
- Exploring Snort (IDS) and pfSense (firewall systems)
- Regular Arch Linux user (3+ years)
- Experience with Docker for containerized services

Hardware

- Designed a custom mechanical keyboard from scratch, including schematic design and PCB layout
- Developed keyboard schematics using Ergogen/Ergopad and implemented the design in KiCad
- Laid out and routed a fully functional PCB, integrating switches, matrix wiring, and supporting components
- Built the keyboard around the ATmega32U4 microcontroller and configured firmware for custom input behavior

Open Source Contributions

- GNU coreutils (**dd**) – contributed via GNU mailing list workflow
- Linux Kernel – documentation contributions via upstream process
- SageMath – contributions to symbolic computation tooling
- Contributor through Linux Foundation, Hacktoberfest, and GSSoC

Projects - IITM

- Vehicle Parking App - multi-user app that manages different parking lots, parking spots and parked vehicles for 4 wheelers. (CS2003P - Modern Application Development I)
- Hospital Management App - Flask and Vue based application for managing hospitals with API, CRUD and database. (CS2006P - Modern Application Development II)

Fellowships

- InSIG (India School on Internet Governance) Fellow – Contributed to discussions and revisions on the Digital Personal Data Protection (DPDP) Act, engaging directly with experts including Rakesh Maheshwari on internet governance policy, in collaboration with the Global Forum on Cyber Expertise (GFCE).
- AuIGF (Australia Internet Governance Forum) Fellow – Participated in global and Australian internet governance discussions, analyzing regulatory challenges and revisiting union laws.
- Youth India Internet Governance Forum (YIGF) – Served on the organizing committee, facilitating sessions and providing insights on emerging trends in the internet governance landscape.

Certificates & Courses

- Virginia Tech and C2QA Quantum Information Science and Engineering Program
- Microsoft SOC Foundations Training
- Quantum Algorithms for Non linear PDEs – Womanium & Wiser Quantum Program 2025
- PennyLane Trotterization Challenge – Womanium & Wiser Quantum Program 2025
- Diploma - QBronze using Qiskit – Womanium & Wiser 2025
- Quantum Computing – CDAC Hyderabad & IIT Roorkee
- OT Security Bootcamp – NFSU
- Quantum computing course, CDAC - Hyderabad and IIT, Roorkee
- (Coursera) Supervised Machine Learning: Regression and Classification - Andrew Ng
- (Coursera) Foundations of Cyber Security - Google
- (AWS) Introduction to Cloud, Semester 1, 2, 3
- (AWS) Machine Learning Foundations

Technical Writing

- Publish technical blogs on Linux, systems programming, cybersecurity, and quantum computing
- Portfolio: sarangvehale.github.io/portfolio

Leadership & Activities

- **Youth India Internet Governance Forum** – *Organizing Committee Member*
 - Collaborated with experts and policymakers to shape the agenda for national discussions on internet governance.
 - Coordinated workshops, panel discussions, and keynotes to promote internet policy awareness across India.
 - Contributed to drafting policy recommendations on digital rights, cybersecurity, and emerging technologies.

- **National Cadet Corps (NCC, Indian Army)** – *Junior Under Officer*
 - Led a squad of cadets in training and drills, fostering leadership and discipline.
 - Represented the unit in inter-school/state-level competitions, earning accolades for team performance.
 - **Camp Senior** – *Army Attachment Camp (5 Sikh Light Infantry unit)*
 - * Managed a high-intensity camp for 540 cadets, overseeing training programs and logistics.
- **All India Forensic Science Conference 2025** – *Organizer, Coordinator, and Liaison Officer*
 - Led planning and execution of a major conference with 200+ professionals in forensic science.
 - Coordinated workshops and panel discussions on criminal justice, law enforcement tech, and emerging trends.
 - Secured sponsorships, managed logistics, and ensured event success.
- **IIT-Bombay Mood India** – *Campus Leader*
 - Led mental health awareness initiatives and organized workshops to reduce stigma.
- **Volunteer** – *Robin Hood Green Army (NGO)*
 - Participated in environmental campaigns, including tree plantations and waste management initiatives.
- **Volunteer & Teacher** – *U&I Foundation (NGO)*
 - Taught underprivileged children, focusing on academics and personal development.

Skills & Interests

Cybersecurity: OSINT, Digital Forensics, Threat Intelligence, MITRE ATT&CK, Incident Response

Quantum Computing: Qiskit, Quantum Algorithms, Quantum Cryptography, Many-Body Systems

Programming: Python, C/C++, Rust, R, Bash, SQL

Tools: Linux, Git, API Integration, Data Scraping

Languages: English, Hindi, Marathi, Sanskrit, Japanese